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1. Toggling with commands in ubuntu to read/access/write files and change ownership

```
echo "Enter file name: "
read file
echo "Enter permission (example 755):"
read perm
chmod $perm $file
echo "Enter new owner name: "
read owner
sudo chown $owner $file
echo "New file permissions are: "
ls -l $file
```

```
ubuntu@ubuntu-OptiPlex-SFF-7020:~/sanjana$ ./ifile.sh
Enter file name:
backgroundtask.sh
Enter permission (example 755):
7
Enter new owner name:
ubuntu
New file permissions are:
-------rwx 1 ubuntu ubuntu 152 Feb 17 15:53 backgroundtask.sh
ubuntu@ubuntu-OptiPlex-SFF-7020:~/sanjana$
```

2. Assigning new owner and new group to a file

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```bash
#!/bin/bash

usage()
{
    echo "Usage: $0 <filename>"
    exit 1
}

if [ "$#" -ne 1 ]; then
    usage
fi

filename=$1

if [ ! -e "$filename" ]; then
    echo "Error: File '$filename' does not exist."
    exit 1
fi

current_permissions=$(ls -l "$filename" | awk '{print $1}')
echo "Current permissions for '$filename': $current_permissions"

read -p "Enter new permissions (e.g., 755 for rwxr-xr-x): " new_permissions

if chmod "$new_permissions" "$filename"; then
    echo "Permissions for '$filename' changed to $new_permissions."
else
    echo "Error: Could not change permissions for '$filename'."
    exit 1
fi
```

[ Read

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execut

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```
GNU nano 7.2                                                    code.sh

if chmod "$new_permissions" "$filename"; then
    echo "Permissions for '$filename' changed to $new_permissions."
else
    echo "Error: Could not change permissions for '$filename'."
    exit 1
fi

new_permissions_display=$(ls -l "$filename" | awk '{print $1}')
echo "New permissions for '$filename': $new_permissions_display"

read -p "Do you want to change the ownership of the file? (y/n): " change_ownership

if [[ "$change_ownership" =~ ^[Yy]$ ]]; then
    read -p "Enter new owner (username): " new_owner
    read -p "Enter new group (groupname): " new_group

    if chown "$new_owner":"$new_group" "$filename"; then
        echo "Ownership for '$filename' changed to $new_owner:$new_group."
    else
        echo "Error: Could not change ownership for '$filename'."
        exit 1
    fi
else
    echo "Ownership change skipped."
fi

echo "Final details for '$filename':"
ls -l "$filename"

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Locati
```

```
ubuntu@ubuntu-OptiPlex-SFF-7020:~/sanjana$ ./code.sh copy.sh
Current permissions for 'copy.sh': -rwxr-xr-x
Enter new permissions (e.g., 755 for rwxr-xr-x): 755
Permissions for 'copy.sh' changed to 755.
New permissions for 'copy.sh': -rwxr-xr-x
Do you want to change the ownership of the file? (y/n): y
Enter new owner (username): ubuntu
Enter new group (groupname):
Ownership for 'copy.sh' changed to ubuntu:.
Final details for 'copy.sh':
-rwxr-xr-x 1 ubuntu ubuntu 233 Feb 24 14:51 copy.sh
ubuntu@ubuntu-OptiPlex-SFF-7020:~/sanjana$
```